

Computer Networks

Local Area Networks

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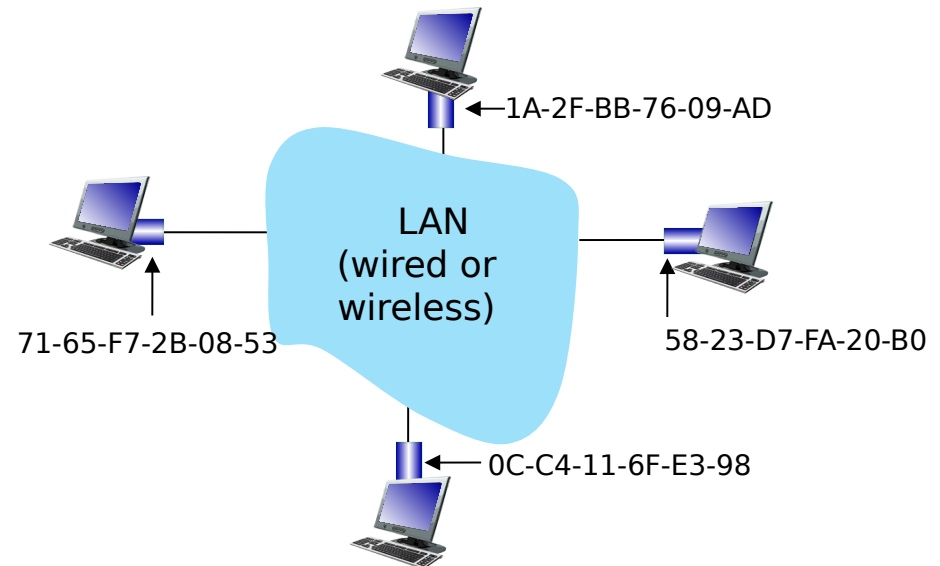
Computer Science and Engineering

IIT Kanpur

MAC Address and Framing

MAC addresses

- MAC (or LAN or physical) address:
 - 48-bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
 - Each interface on LAN has an **unique** MAC address
 - e.g.: 1A-2F-BB-76-09-AD



MAC addresses

- MAC address allocation administered by IEEE
 - Manufacturer buys portion of MAC address space (to assure uniqueness)
 - MAC address: portability
 - Can move interface from one LAN to another
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Ethernet frame structure

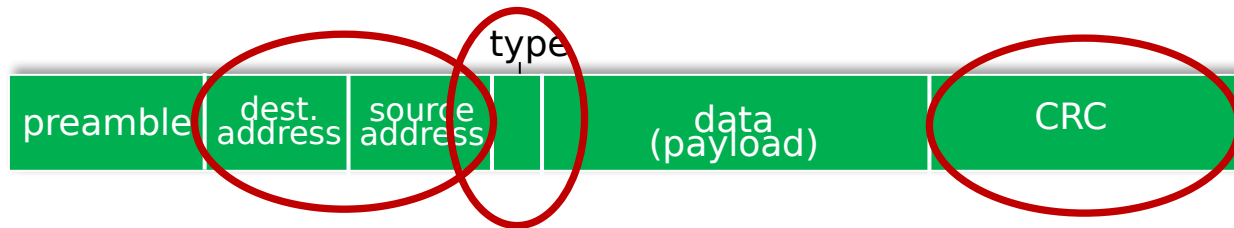
Sending interface encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



Preamble:

- Used to synchronize receiver, sender clock rates
 - 7 bytes of 10101010 followed by one byte of 10101011
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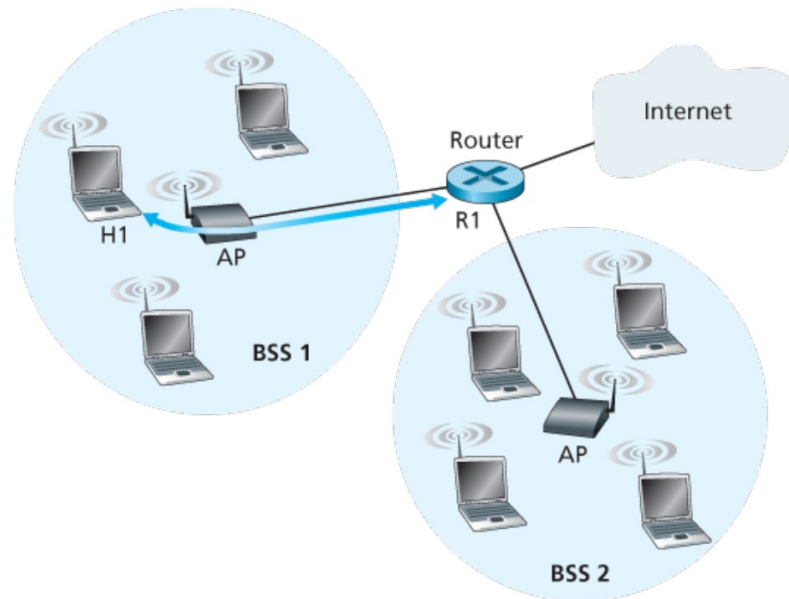
Ethernet frame structure



- **Addresses:** 6 byte source, destination MAC addresses
 - If adapter receives frame with matching destination address, or with broadcast address, it passes data in frame to network layer protocol
 - Otherwise, adapter discards frame
 - **Type:** indicates higher layer protocol
 - Mostly IP but others possible, e.g., Novell IPX, AppleTalk
 - Used to demultiplex up at receiver
 - **CRC:** Cyclic redundancy check at receiver
 - Error detected: frame is dropped
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802.11 frame structure

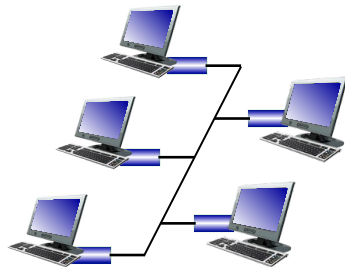
Bytes	2	2	6	6	6	2	0-2312	4
	Frame control	Duration	Address 1 (recipient)	Address 2 (transmitter)	Address 3	Sequence	Data	Check sequence



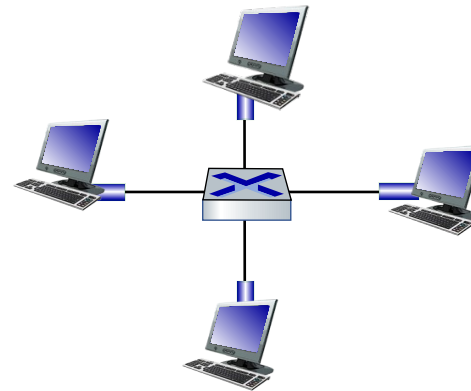
Ethernet Switches

Ethernet: physical topology

- **Bus:** popular through mid 90s
 - All nodes in same collision domain (can collide with each other)
- **Switched:** prevails today
 - Active link-layer 2 **switch** in center
 - Each “spoke” runs a (separate) Ethernet protocol (nodes do not collide with each other)



Bus



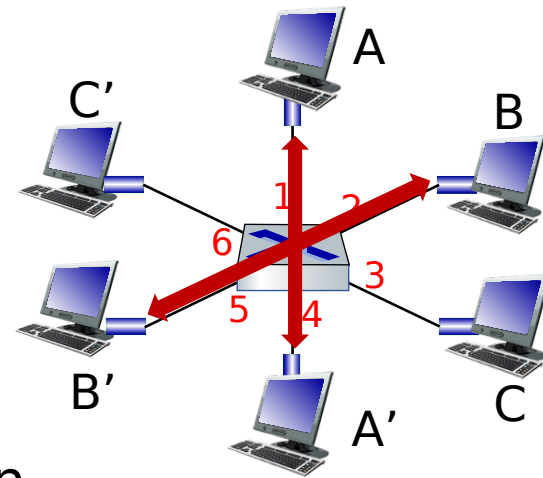
Switched

Ethernet switch

- Switch is a **link-layer** device
 - Store, forward Ethernet frames
 - Examine incoming frame's MAC address, **selectively** forward frame to one-or-more outgoing links when frame is to be forwarded on segment, uses CSMA/CD to access segment
 - **Transparent:** hosts unaware of presence of switches
 - **Plug-and-play, self-learning**
 - Switches do not need to be configured
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Ethernet switch

- Hosts have dedicated, direct connection to switch
- Switches buffer packets
- Ethernet protocol used on each incoming link, so:
 - No collisions; full duplex
 - Each link is its own collision domain
- **Switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions



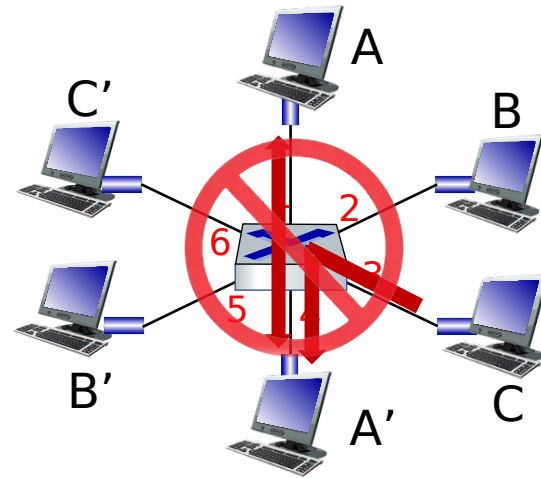
switch with six
interfaces

(1 2 3 4 5 6)

<https://www.ciscopress.com/articles/article.asp?p=2111396>

Ethernet switch

- Hosts have dedicated, direct connection to switch
- Switches buffer packets
- Ethernet protocol used on each incoming link, so:
 - No collisions; full duplex
 - Each link is its own collision domain
- **Switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions
 - But A-to-A' and C to A' can not happen simultaneously



switch with six
interfaces
(1,2,3,4,5,6)

<https://study-ccna.com/half-duplex-and-full-duplex/>

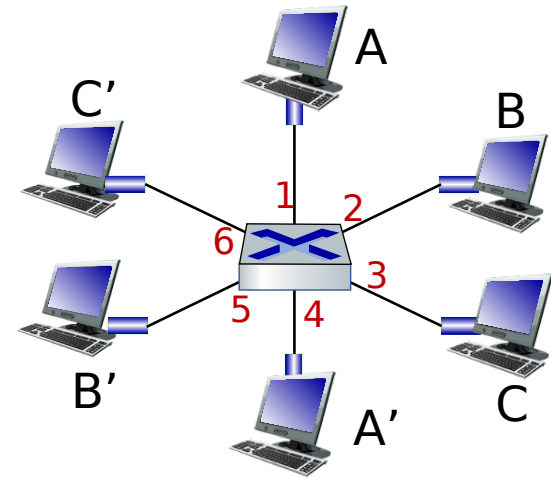
Switch forwarding table

Q: How does switch know A' reachable via interface 4, B' reachable via interface 5?

A: Each switch has a **forwarding table**, each entry:

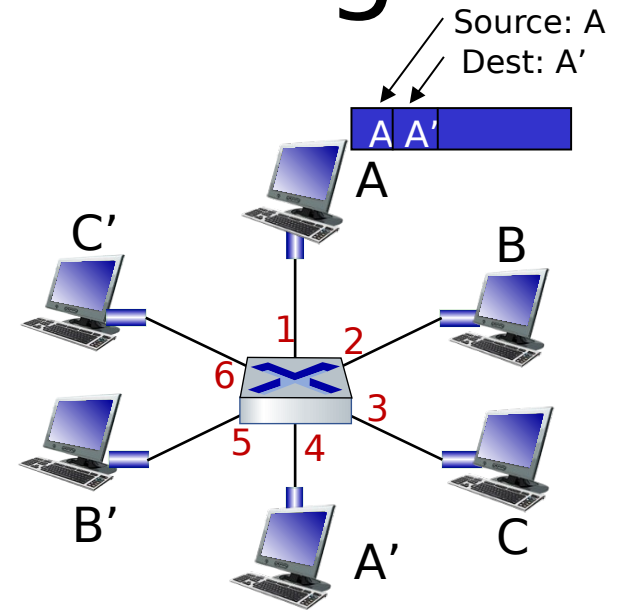
- (MAC address of host, interface to reach host, time stamp)

Q: How are entries created, maintained in forwarding table?



Switch: self-learning

- Switch **learns** which hosts can be reached through which interfaces
 - When frames are received, switch “learns” location of sender: incoming LAN segment
 - Records sender/location pair in switch table

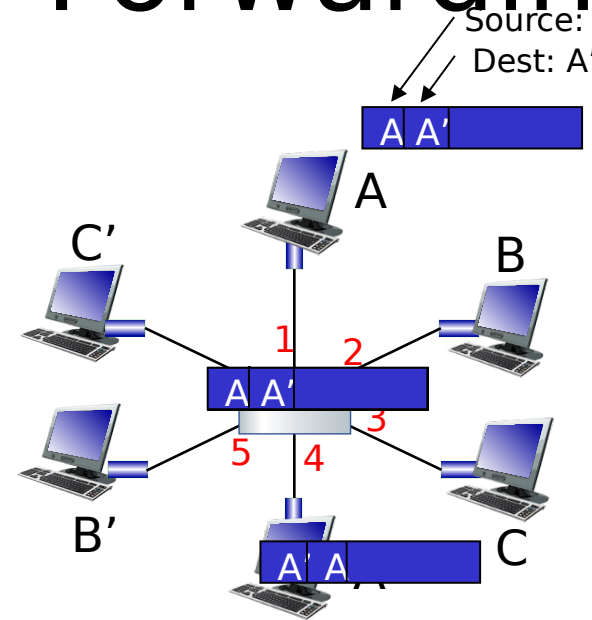


MAC addr	interface	TTL
A	1	60

Forwarding table
(initially empty)

Self-learning and Forwarding

- Frame destination, A', location unknown:
flood
- Destination A location known:
selectively send
on just one link

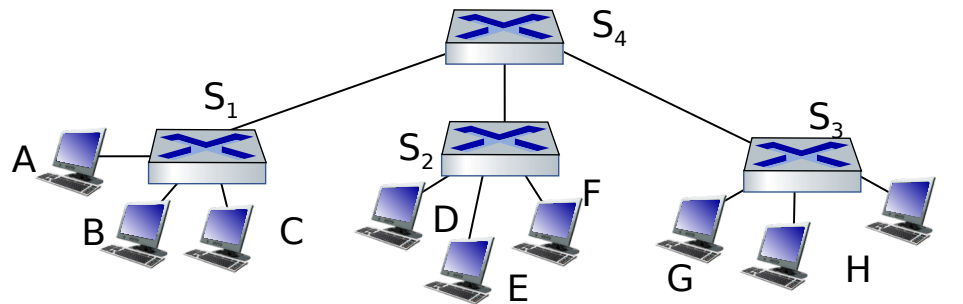


MAC addr	interface	TTL
A	1	60
A'	4	60

switch table
(initially empty)

Interconnecting switches

Self-learning switches can be connected together:



- Q:** Sending from A to G - how does S₁ know to forward frame destined to G via S₄ and S₃?
- **A:** Self learning! (works exactly the same as in single-switch case!)
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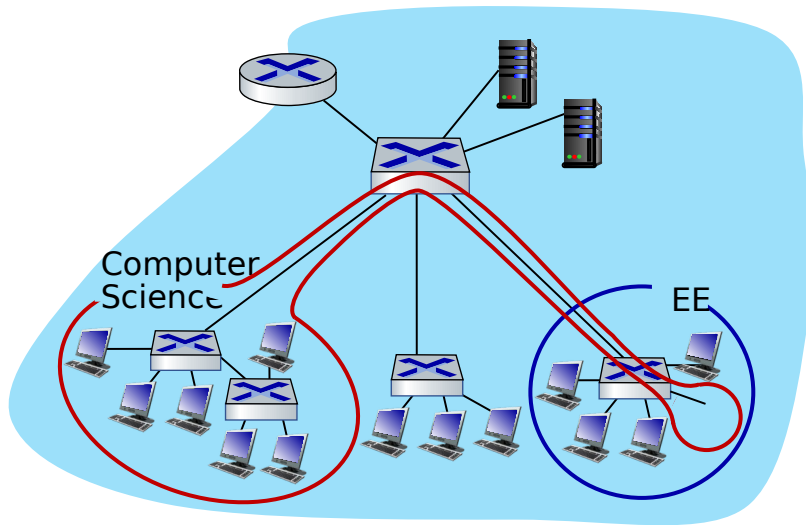
Switch: frame filtering/forwarding

When frame received at switch:

1. Record incoming link, MAC address of sending host
 2. Index forwarding table using MAC destination address
 3. If entry found for destination
then {
 if destination on segment from which frame arrived
 then drop frame
 else forward frame on interface indicated by entry
}
else flood /* forward on all interfaces except arriving interface */
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Virtual LAN

Virtual LANs (VLANs)



Single broadcast domain:

- **Scaling:** all layer-2 broadcast traffic (i.e. unknown MAC) must cross entire LAN
- Efficiency, security, privacy issues

Administrative issues:

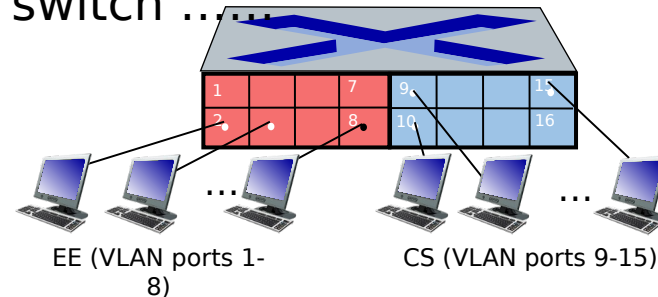
- CS user moves office to EE - **physically** attached to EE switch, but wants to remain **logically** attached to CS switch

Port-based VLANs

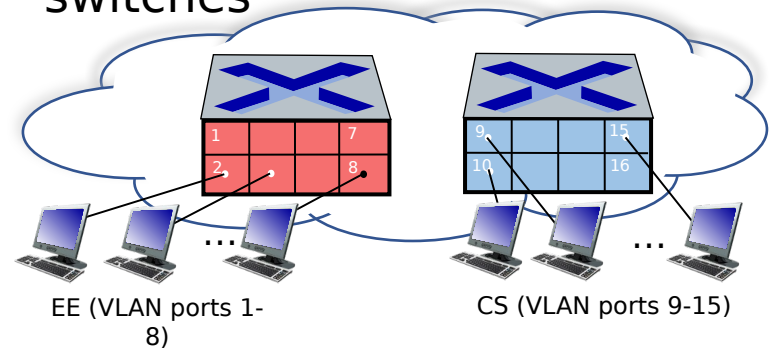
Virtual Local Area Network (VLAN)

Switches supporting VLAN capabilities can be configured to define multiple **virtual** LANs over single physical LAN infrastructure

Port-based VLAN: switch ports grouped (by switch management software) so that **single** physical switch

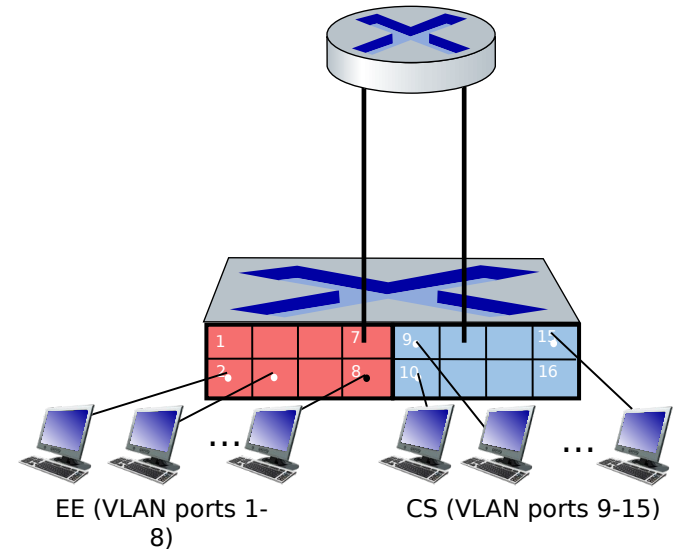


... operates as **multiple** virtual switches

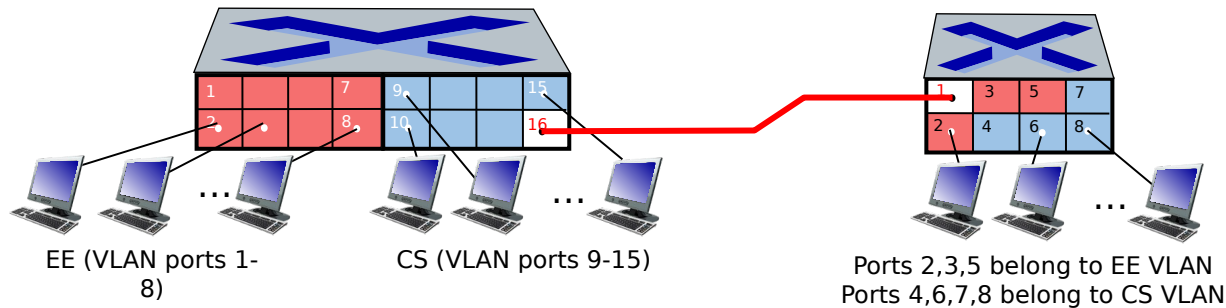


Port-based VLANs

- **Traffic isolation:** frames to/from ports 1-8 can only reach ports 1-8
 - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- **Dynamic membership:** ports can be dynamically assigned among VLANs
- **Forwarding between VLANs:** done via routing (just as with separate switches)
 - in practice vendors sell combined switches plus routers



VLANs spanning multiple switches



Trunk port: carries frames between VLANs defined over multiple physical switches

- frames forwarded within VLAN between switches must carry VLAN ID info
- 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports

Summary

□ Local area networks:

- MAC addresses and framing
 - Ethernet switch and self learning
 - Virtual LANs
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